# Paper Review Task 1

## Literature Review on

# Deep Learning Approaches for Brain Tumour Analysis

Submitted by,

Name: Tasnuba Tasnim
ID: 2671508
Institute of Information and Communication Technology
Khulna University of Engineering and Technology.

Submitted to,

Dr. M. A. Moyeen
Assistant Professor
Institute of Information and Communication Technology
Khulna University of Engineering and Technology.

# Table of content

Abstract

This paper review assignment examines the 2026 systematic review by Tariq et al. on deep-learning approaches for brain tumour detection and diagnosis. The review synthesizes evidence across segmentation, detection, classification, grading and emerging prognostic applications. Convolutional neural networks remain the foundational model family, while attention mechanisms, transformers, graph networks and hybrid systems are increasingly used to improve global context, boundary delineation and robustness. The main conclusion is that high benchmark performance alone is insufficient for precision oriented clinical use. Reliable deployment requires external validation, calibrated uncertainty, explainability, privacy preserving collaboration, demographic fairness, regulatory readiness and longitudinal evaluation. This document summarizes the paper's problem statement, taxonomy, security and privacy discussion, open challenges and limitations, and provides a critical analysis and a comparative table of representative references cited by the review.

# Selected Paper and Review Objective

The selected paper is Systematic Review for Detecting & Diagnosing Brain Tumor Using Deep Learning: Current Trends and Future Research Directions, published by Springer Nature in the International Journal of Computational Intelligence Systems in March 2026 [1]. The paper aims to organize recent evidence on automated brain-tumour analysis, compare model families and validation practices, and identify barriers that prevent research prototypes from becoming dependable clinical decision-support tools.

The review follows a PRISMA-P-guided protocol, prospectively registered in PROSPERO (CRD420251267246), and uses QUADAS-2 to assess methodological quality and risk of bias [1]. Data extraction covers dataset characteristics, label provenance, preprocessing, augmentation, architecture, objectives, validation strategy, performance metrics and interpretability. This methodological breadth is a major strength because it evaluates not only predictive performance but also reproducibility, generalization, governance and clinical translation.

# Problem Statement

Brain tumours are highly heterogeneous: their size, location, boundaries, tissue characteristics and biological behaviour vary across patients and tumour subtypes. Manual interpretation and tumour outlining are labour intensive and affected by inter observer variation. Deep learning models can automate important parts of detection, segmentation and classification, but the literature reports inconsistent performance across datasets, institutions, scanners, patient groups and annotation practices [1].

The central problem is therefore not simply whether a CNN can achieve high accuracy on a benchmark. The more important question is whether the model can remain reliable on unseen patients, rare tumour types, small or ambiguous lesions and data produced at different hospitals. For precision medicine, the model must also move beyond tumour presence toward patient specific characterisation, grading, molecular or prognostic inference, treatment response assessment and longitudinal monitoring.

# Review Methodology

- Protocol and transparency: PRISMA-P guided planning with prospective PROSPERO registration.
- Search and selection: predefined eligibility criteria, duplicate removal, title/abstract screening and full-text review.
- Quality assessment: QUADAS-2 evaluation of patient selection, index test, reference standard, flow and timing.
- Extraction variables: data source, cohort, tumour type, preprocessing, augmentation, architecture, training objective, validation, performance and interpretability.
- Synthesis: structured qualitative comparison because substantial study heterogeneity prevented a formal meta-analysis.

The review reports that the literature is dominated by segmentation and classification/grading, while pure detection and patient level diagnostic or prognostic modelling receive less attention. Segmentation accounts for approximately 36.4% of the reviewed research, classification/grading 32.6%, detection 18.5%, and diagnostic/prognostic modelling only 3.3% [1]. This distribution is important: it shows that the field is technically mature in image level tasks but still relatively weak in the outcomes most directly connected to precision medicine.

# Taxonomy of Existing Solutions

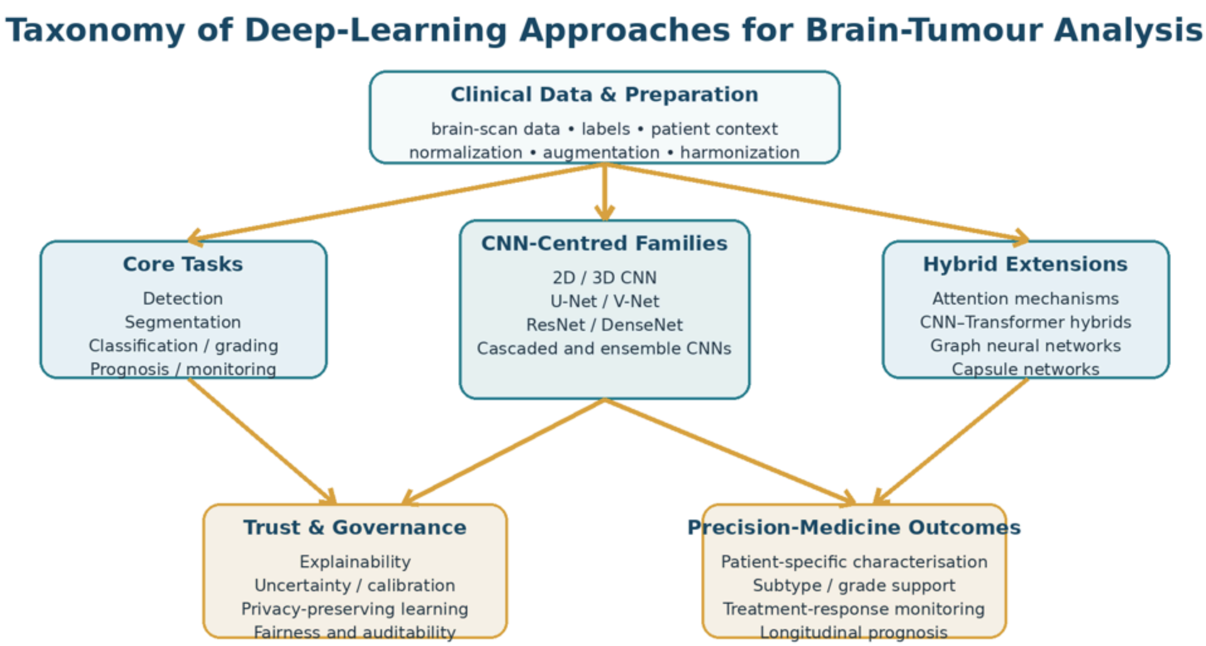


Figure 1. Assignment taxonomy synthesized from the model families and translation themes in the selected review [1].

## CNN-Centred Architectures

CNNs are the backbone of the reviewed literature. Encoder decoder networks such as U-Net and V-Net dominate tumour segmentation, while ResNet, DenseNet, VGG and lightweight custom CNNs are widely used for detection, classification and grading. Three dimensional CNNs capture volumetric context more effectively than two dimensional models, but they require more memory, computation and training data. The review reports approximately 0.89 whole tumour Dice for an nnU-Net ensemble on BraTS 2020, while representative CNN classifiers achieved roughly 91–97% accuracy on curated classification tasks [1–3].

## Hybrid and Emerging Architectures

Hybrid systems extend CNNs with attention, transformers, graph learning or capsule structures. CNN Transformer models combine local feature extraction with long-range contextual modelling; the review cites TransBTS-type systems with about 0.91 whole-tumour Dice on BraTS 2019 [1]. Graph models represent relational structure among regions, while capsule networks attempt to preserve part whole relationships and rotational information. These methods can improve difficult cases but usually increase computational cost, optimization complexity and data requirements.

## Algorithmic Enhancements

Common enhancements include Dice and focal losses for class imbalance, deep supervision, patch based learning, extensive augmentation, cascaded refinement, ensemble prediction, post processing and uncertainty estimation. These techniques can improve small lesion performance and stability, but the review warns that a preprocessing or optimization step may improve one metric while degrading another. Rigorous ablation studies are therefore essential.

# Connection to Precision Medicine

The review is primarily organized around automated tumour analysis tasks, but several themes connect it to precision medicine. Accurate segmentation can quantify patient specific tumour burden; classification and grading can support subtype aware decisions; integration of clinical, genomic or molecular information can help predict recurrence, therapy resistance and prognosis; and longitudinal models can monitor progression and treatment response. The review specifically identifies serial-data modelling, radiogenomic integration, multi-task learning and patient specific prognosis as important future directions [1, 12].

However, the precision-medicine component remains less developed than the segmentation and classification literature. Only a small share of the reviewed studies focuses on diagnostic/prognostic modelling. Consequently, the strongest interpretation is that current CNN systems provide enabling components for precision medicine rather than complete personalized treatment systems.

# Security, Privacy and Ethical AI

Brain-tumour datasets contain sensitive patient information, and multi centre collaboration is often needed to obtain sufficient diversity. The review therefore treats privacy, fairness and governance as part of technical model quality rather than as optional legal add-ons [1].

1. De-identification and secure storage: Patient identifiers should be removed, sensitive metadata masked, access controlled, and data encrypted both in transit and at rest. Audit trails are needed to record access and modification events.
2. Federated learning: Hospitals train local models and share parameters or gradients rather than raw patient data. This supports data sovereignty and allows collaboration across institutions.
3. Secure aggregation and multiparty computation. Cryptographic protocols can prevent participants or a central server from seeing individual institutional updates.
4. Differential privacy: Noise can be added to model updates to reduce re-identification or membership inference risk, although excessive noise may reduce accuracy, especially for rare tumour patterns.
5. Fairness aware optimization: A large institution or majority demographic can dominate collaborative training. Reweighting, subgroup evaluation and adversarial fairness constraints can reduce unequal error rates.
6. Explainability and human oversight: Grad-CAM, attention maps and similar tools can indicate influential regions, but explanations must be clinically validated. Human-in-the-loop review should remain central to high-stakes decisions.

The major unresolved security issue is the privacy utility fairness trade-off. Federated learning limits raw data sharing, but model updates can still leak information; privacy mechanisms can reduce this risk but may reduce performance; and a privacy preserving model can still be biased. Security, fairness and clinical validation therefore have to be evaluated together rather than independently.

# Open Challenges Identified by the Review

1. Cross-site generalization: Models often lose performance when transferred to unseen institutions, patient populations or acquisition protocols. The review reports external validation Dice reductions of roughly 8–15% for some single-centre models.
2. Ambiguous boundaries and small lesions: Even high overlap models may produce clinically important boundary errors. Small enhancing regions and infiltrative margins remain difficult.
3. Class imbalance and rare tumours: Public datasets are dominated by common adult gliomas. Pediatric and rare tumour types are underrepresented, creating unequal sensitivity and weak generalization.
4. Dataset and annotation heterogeneity: Different labels, preprocessing pipelines, splits and metrics make results difficult to compare. Data leakage and patient level split errors can inflate reported performance.
5. Calibration and uncertainty: Accuracy and Dice do not show whether confidence is trustworthy. Uncertainty reporting, calibration and failure detection remain incomplete.

6. Explainability and clinical trust: Heatmaps can be visually persuasive without being clinically meaningful. Explanations need stability tests, causal interpretation and expert evaluation.
7. Regulatory and workflow integration: Clinical adoption requires prospective trials, lifecycle controls, cybersecurity, interoperability and usable integration with established workflows.
8. Longitudinal precision outcomes: Most studies analyse a single time point. Patient specific prognosis, treatment response prediction and adaptive monitoring need curated longitudinal and multimodal data.

# Critical Analysis

## Strengths

1. The review uses a registered, PRISMA-guided protocol and QUADAS-2 quality assessment.
2. It moves beyond architecture accuracy to include external validation, reproducibility, privacy, fairness, regulation and deployment.
3. Its taxonomy clearly shows the transition from conventional CNNs to attention, transformers, graph models and hybrid systems.
4. The review gives special attention to pediatric underrepresentation, domain shift and privacy preserving collaboration.

## Limitations and Missing Perspectives

1. The evidence base is restricted mainly to English-language studies published from 2019 to 2025, which may omit relevant earlier, non English or unpublished evidence.
2. Study heterogeneity prevents a formal meta-analysis, so many comparisons remain narrative rather than statistically pooled.
3. Heavy reliance on the BraTS benchmark can create benchmark-specific optimism and weak real-world generalization.
4. The paper reports 81 included studies in one section and 82 in another, an editorial inconsistency that reduces confidence in corpus accounting.
5. Although privacy methods are discussed, the paper provides limited empirical comparison of privacy budgets, communication cost, security threats and accuracy loss.
6. The paper is stronger on segmentation and classification than on personalized prognosis, molecular inference, treatment selection and patient outcomes.
7. Explainability is presented mainly through common visualization methods; causal validity, explanation stability and clinician-centred evaluation need deeper treatment.

## Personal Evaluation

In my assessment, this is a strong source for the assignment because it covers all required elements: a clear healthcare problem, a usable taxonomy, privacy and security mechanisms, open challenges and clinical limitations. Its greatest contribution is the argument that clinical reliability depends on more than benchmark accuracy. Nevertheless, the phrase “precision medicine” should be used carefully. The reviewed literature mainly develops technical components for personalized care, while only a small portion directly predicts patient outcomes or treatment response. A more evidence aligned framing is therefore “deep learning approaches for brain-tumour analysis and precision oriented clinical decision support.”

## Conclusion

Deep learning has advanced automated brain-tumour analysis, with CNNs remaining central to segmentation, detection, classification and grading. Hybrid attention, transformer and graph based models improve contextual modelling, but their value is constrained by computational cost, dataset bias and weak external validation. Future work must prioritize cross-site robustness, pediatric and rare-tumour representation, calibrated uncertainty, explainability, privacy, fairness, regulation and longitudinal outcomes.

## Single Page Comparative Summary of Representative References

| **Reference** | **Primary task** | **Approach** | **Reported contribution/result** | **Precision or clinical relevance** | **Main limitation** |
|---|---|---|---|---|---|
| Isensee et al. (2021) [2] | Segmentation | Auto-configured 3D U-Net ensemble (nnU-Net) | Review reports ~0.89 whole-tumour Dice on BraTS 2020. | Strong reproducible baseline for tumour-volume measurement. | High compute; benchmark dependence; external transfer remains uncertain. |
| Lin et al. (2023) [5] | Segmentation | CNN Transformer with clinical knowledge cross-attention | Combines local convolutional features and long-range context. | Improves structured tumour-region characterisation. | Complex training and large data/compute requirements. |
| Goceri (2024) [6] | Classification /grading | Efficient CNN and transformer blocks | Targets tumour classification and glioma grading in one architecture. | Supports subtype- and grade-aware decisions. | Needs broad multi-centre and prospective validation. |
| Dewage et al. (2024) [7] | Detection | Custom CNN with interpretability analysis | Links detection performance with explanation-oriented evaluation. | Can improve clinician review of model focus. | Interpretability does not prove causal or clinical correctness. |
| M. et al. (2024) [8] | Detection / classification | ResNet-50 with Grad-CAM | Produces visual explanations alongside predictions. | Supports human-in-the-loop checking. | Heatmaps may be unstable and dataset-specific. |
| Choudhry et al. (2024) [9] | Segmentation | Interpretable graph convolutional network | Uses relational structure beyond regular convolution. | May represent complex tumour-region interactions. | Higher architectural complexity; evidence remains preliminary. |
| Lata et al. (2024) [10] | Privacy-preserving detection | Deep learning in privacy-preserving smart healthcare | Demonstrates security-aware tumour-analysis design. | Enables collaborative learning with less raw-data exposure. | Privacy, communication and accuracy trade-offs require testing. |

| | | | | | |
|---|---|---|---|---|---|
| Cariola et al. (2025) [11] | Pediatric segmentation | Attention U-Net ensemble and post-processing | Review reports enhancing-tumour Dice improvement from 0.757 to 0.764. | Addresses an underrepresented patient population. | Small/rare cohorts and limited generalization. |
| Chhotray et al. (2025) [13] | Classification | Optimized cascaded CNN | Refines feature extraction through staged processing. | Potentially improves difficult subtype discrimination. | Cascades increase inference and maintenance complexity. |
| Aziz et al. (2024) [14] | Classification | Pretrained DenseNet, global pooling and tuning | Emphasizes both precision and generalization. | Transfer learning helps when labelled data are limited. | Public-dataset performance may not reflect clinical diversity. |
| Mathivanan et al. (2024) [15] | Detection | Deep learning and transfer learning | Reports accurate tumour detection using pretrained models. | Data-efficient baseline for deployment studies. | External validation and subgroup performance need stronger evidence. |
| Khalighi et al. (2024) [12] | Diagnosis / prognosis / treatment | Review of AI in neuro-oncology | Connects AI analysis with prognosis and precision treatment. | Directly frames patient-specific clinical use. | Broader than CNNs; translational evidence remains limited. |

Note: Outcomes are summarized from the selected systematic review; reported values are study and dataset-specific and should not be interpreted as directly comparable clinical performance.

# References

[1] U. Tariq, I. Ahmed, K. Shaukat, M. Ihsan, et al., "Systematic Review for Detecting & Diagnosing Brain Tumor Using Deep Learning: Current Trends and Future Research Directions," International Journal of Computational Intelligence Systems, vol. 19, art. 189, 2026. doi: 10.1007/s44196-026-01283-2.

[2] F. Isensee, P. F. Jäger, P. M. Full, P. Vollmuth, and K. H. Maier-Hein, "nnU-Net for Brain Tumor Segmentation," Lecture Notes in Computer Science, pp. 118–132, 2021. doi: 10.1007/978-3-030-72087-2_11.

[3] T.-W. Wang et al., "Artificial Intelligence Detection and Segmentation Models: A Systematic Review and Meta-Analysis of Brain Tumors in Magnetic Resonance Imaging," Mayo Clinic Proceedings: Digital Health, vol. 2, no. 1, pp. 75–91, 2024. doi: 10.1016/j.mcpdig.2024.01.002.

[4] S. Solanki, U. P. Singh, S. S. Chouhan, and S. Jain, "Brain Tumor Detection and Classification Using Intelligence Techniques: An Overview," IEEE Access, vol. 11, pp. 12870–12886, 2023. doi: 10.1109/ACCESS.2023.3242666.

[5] J. Lin et al., "CKD-TransBTS: Clinical Knowledge-Driven Hybrid Transformer with Modality-Correlated Cross-Attention for Brain Tumor Segmentation," IEEE Transactions on Medical Imaging, vol. 42, no. 8, pp. 2451–2461, 2023. doi: 10.1109/TMI.2023.3250474.

[6] E. Goceri, "An Efficient Network with CNN and Transformer Blocks for Glioma Grading and Brain Tumor Classification," Expert Systems with Applications, vol. 268, art. 126290, 2024. doi: 10.1016/j.eswa.2024.126290.

[7] K. A. K. W. Dewage, R. Hasan, B. Rehman, and S. Mahmood, "Enhancing Brain Tumor Detection through Custom Convolutional Neural Networks and Interpretability-Driven Analysis," Information, vol. 15, no. 10, art. 653, 2024. doi: 10.3390/info15100653.

[8] M. M., M. T. R., V. V. Kumar, and S. Guluwadi, "Enhancing Brain Tumor Detection through Explainable AI Using Grad-CAM with ResNet-50," BMC Medical Imaging, vol. 24, 2024. doi: 10.1186/s12880-024-01292-7.

[9] A. Choudhry, S. Iqbal, M. Alhussein, K. Aurangzeb, A. N. Qureshi, and A. Hussain, "A Novel Interpretable Graph Convolutional Neural Network for Multimodal Brain Tumor Segmentation," Cognitive Computation, vol. 17, 2024. doi: 10.1007/s12559-024-10387-w.

[10] K. Lata, P. Singh, S. Saini, and L. R. Cenkeramaddi, "Deep Learning-Based Brain Tumor Detection in Privacy-Preserving Smart Health Care Systems," IEEE Access, vol. 12, pp. 140722–140733, 2024. doi: 10.1109/ACCESS.2024.3456599.

[11] E. Cariola, A. Sibilano, V. Guerriero, A. Bevilacqua, and A. Brunetti, "Deep Learning Strategies for Semantic Segmentation of Pediatric Brain Tumors," Scientific Reports, vol. 15, 2025. doi: 10.1038/s41598-025-07257-2.

[12] S. Khalighi, K. Reddy, A. Midya, K. B. Pandav, A. Madabhushi, and M. Abedalthagafi, "Artificial Intelligence in Neuro-Oncology: Advances and Challenges in Brain Tumor Diagnosis, Prognosis, and Precision Treatment," npj Precision Oncology, vol. 8, 2024. doi: 10.1038/s41698-024-00575-0.

[13] S. K. Chhotray, D. Mishra, S. P. Pati, and S. Mishra, “An Optimized Cascaded CNN Approach for Feature Extraction from Brain MRIs for Tumor Classification,” IEEE Access, vol. 13, pp. 32681–32705, 2025. doi: 10.1109/ACCESS.2025.3543214.

[14] N. Aziz, N. Minallah, J. Frnda, M. Sher, M. Zeeshan, and A. H. Durrani, “Precision Meets Generalization: Enhancing Brain Tumor Classification via Pretrained DenseNet with Global Average Pooling and Hyperparameter Tuning,” PLOS ONE, vol. 19, no. 9, e0307825, 2024. doi: 10.1371/journal.pone.0307825.

[15] S. K. Mathivanan, S. Sonaimuthu, S. Murugesan, H. Rajadurai, B. D. Shivahare, and M. A. Shah, “Employing Deep Learning and Transfer Learning for Accurate Brain Tumor Detection,” Scientific Reports, vol. 14, 2024. doi: 10.1038/s41598-024-57970-7.